LIQUEFIED GASES IN BULK (IGC CODE) - 2 _
ANNEX
AMENDMENTS TO THE INTERNATIONAL CODE FOR THE CONSTRUCTION AND EQUIPMENT OF SHIPS CARRYING LIQUEFIED GASES IN BULK (IGC CODE)
Amendments related to application
1 Existing paragraphs 1.1.2 and 1.1.3 are repla ce d by the following:
"1.1. 2 Unless expressly provided otherwise, the Code applies. to s hips the keels of which are laid or which are at a stage at whi ch:

---CHUNK---

SOURCE: MSC.32(63).pdf
Amendments related to application
1 Existing paragraphs 1.1.2 and 1.1.3 are repla ce d by the following:
"1.1. 2 Unless expressly provided otherwise, the Code applies. to s hips the keels of which are laid or which are at a stage at whi ch:
.1 construction identifiable with the ship begins; and
.2 assembly of that ship has commenced comprising at least 50 teunes or l' of the estimated mass of all structural mate rial, whichever is the less;
on or after 1 July 1998. Ships constructed before 1 July 1998 are to comply with resolution MSC.5(48) adopted on 1 7 June 1963 subject to amendment s by r e solution MSC.30(61) adopted on 11 December 1992.
1.1.3 A ship. irrespective of the date of construction, which is converted to a gas carrier on or after 1 July 1998, should be treated as a gas carrier constructed on the date on which such conversion commences."
Amendments related to fillipg limits
2 The existing chapter 15 is replaced by the following:
"CHAPTER 15
FILLING LIMITS FOR CARGO TANKS
15.1 General
15.1.1 No cargo tanks should have a higher filling limit (FL) than 98' at the reference temperature, except as permitted by 15.1. 3.

---CHUNK---

SOURCE: MSC.32(63).pdf
"CHAPTER 15
FILLING LIMITS FOR CARGO TANKS
15.1 General
15.1.1 No cargo tanks should have a higher filling limit (FL) than 98' at the reference temperature, except as permitted by 15.1. 3.
15.1.2 The maximum loading limit (LL) to which a cargo tank may be loaded should be determined by the following formula:
LL
where:
LL
FL
PR
FL
PL
loading limit exp r essed as a pe r centage, being the maximum allowable liguid volume rela tive to the tank volume to which the tank may be loaded;

filling limit as specified i n 15.1.1 or 15.1.3;
RESOLUTION MSC.32(63)
adopted on 23 May 1994
ADOPTION OF AMENDMENTS TO THE INTERNATIONAL CODE FOR
THE CONSTRUCTION AND EQUIPMENT OF SHIPS CARRYING
LIQUEFIED GASES IN BULK (IGC CODE) PR
PL
~ 3 ~
relative density of cargo at the reference
temperature: and
relative density of cargo at the loading temperature
and pressure.
15.1.3 The Administration may allow a higher f illing limit (FL) than the
limit of 98' specified in 15.1.1 at the reference temperature, taking int o
account the shape of the tank, arrangements of pressure relief valves,
accuracy of level and temperature gauging and the difference between the

---CHUNK---

SOURCE: MSC.32(63).pdf
limit of 98' specified in 15.1.1 at the reference temperature, taking int o
account the shape of the tank, arrangements of pressure relief valves,
accuracy of level and temperature gauging and the difference between the
loading temperature and the temperature corresponding to the vapour pressure
of the cargo at the set pressure of the pressure relief valves, provided the
conditions specified in 8.2.17 are maintained.
15.1.4 For the purposes of this chapter only, "reference temperature" means:
.1 the temperature corresponding to the vapour pressure of the cargo at
the set pressure of the pressure relief valves when DO cargo vapour
pressure/temperature control as referred to in chapter 7 is provided;
.2 the temperature of the cargo upon termination of loading, during
transport. or at unloading, whichever is the greatest. when a cargo
vapour pressure/temperature control as referred to in chapter 7 is
provided. If this reference temperature would result in the cargo
tank becoming liquid full before the cargo reaches a temperature
corresponding to the vapour pressure of the cargo at the set
pressure of the relief valves required in 8.2. an additional

---CHUNK---

SOURCE: MSC.32(63).pdf
tank becoming liquid full before the cargo reaches a temperature
corresponding to the vapour pressure of the cargo at the set
pressure of the relief valves required in 8.2. an additional
pressure relieving system complying with 8.3 should be fitted.
15.1.5 The Administration may allow type C tanks to be loaded acco r ding
to the following formula, provided that the tank vent system has been appr oved
in accordance with 8.2.18:
LL
where:
LL
FL
PL
PR

Fe
PL
loading limit as specified in 15.1.2;
filling limit as specified in 15.1.1 or 15.1.3;
relative density of cargo at the highest temperature
which the cargo may reach upon termination of loading,
during transport, or at unloading. under the ambient
design temperature conditions described in 7.1.2; and
as specif i ed in 15.1.2.
This paragraph does not apply to products requiring a type IG ship.
RESOLUTION MSC.32(63)
adopted on 23 May 1994
ADOPTION OF AMENDMENTS TO THE INTERNATIONAL CODE FOR
THE CONSTRUCTION AND EQUIPMENT OF SHIPS CARRYING
LIQUEFIED GASES IN BULK (IGC CODE) - 4 -
15.2 Information to be provided to the master
The maximum a l lowable loading limits for each cargo tank should be

---CHUNK---

SOURCE: MSC.32(63).pdf
THE CONSTRUCTION AND EQUIPMENT OF SHIPS CARRYING
LIQUEFIED GASES IN BULK (IGC CODE) - 4 -
15.2 Information to be provided to the master
The maximum a l lowable loading limits for each cargo tank should be
indicat~d for each product which may be carried, for each load ing temperature
which may be applied and for the applicable maximum reference tem per a t ure, on
a list to be approved by the Administration. Pressures at which the pressure
relief valves, including those valves required by 8.3, have been set should
also be stated on the list. A copy of the li st should be kep t permanently on
board by the master.
15.3 Chapter 15 app lie s to all ships regardless of the date of const ruction. "
The fol l owing words are added at the end o f ex i sti ng pa ragraph 8 . 2.17:
"at the ma:z:imum allowable filling limit (FL)",
4 The following new paragraph 8.2.18 i s added after exis ting
paragraph 8.2.17:
"8.2.18 The adeguacy of the vent system fitted on tank s l oaded in
accordance with 15.1 . 5 is to be demonstrated using the guidelines
developed by the Organization. A relevant certificate should be

---CHUNK---

SOURCE: MSC.32(63).pdf
paragraph 8.2.17:
"8.2.18 The adeguacy of the vent system fitted on tank s l oaded in
accordance with 15.1 . 5 is to be demonstrated using the guidelines
developed by the Organization. A relevant certificate should be
permanently kept on board the ship. For the purposes of this pa ragraph.
vent system means:
.1 the tank outlet and the piping to the pressure relief valve;
.2 the pressure relief valve;
.3 the piping fro m the pressure relief valve to the location of
discharge to the atmosphere and including any interconnections
and piping which joins other tank s .

This paragraph may apply to all ships rega rdl es s of t he date of construction. "

Amendments r e lated to cargg tank vent system s

5 The existing paragraph 8.2.3 is replaced by the foll owin g :

"8.2.3 In general, the setting of the pressure relief valves should not be higher than the vapour pressure which ha s been used in t he design of the tank. However. where two or more pressure relief valves are fitted, va l ves comprising not more than 5o, of the total relieving capa city may be set at a pressure up to 5'\, above HARVS."

6 The following sentences are added to existing paragraph 8.2.4:

---CHUNK---

SOURCE: MSC.32(63).pdf

va l ves comprising not more than 5o, of the total relieving capa city may be set at a pressure up to 5'\, above HARVS."

6 The following sentences are added to existing paragraph 8.2.4:

"Valves should be constructed of materia ls with a melti ng point above 925°C. Consideration should be given to lower melting point materials for internal parts and seal s if their use wi ll yield a significant improvement in the general operation of the valve."

RESOLUTION MSC.32(63)

adopted on 23 May 1994